



CKA EXAM PREP

CKA Question Bank

2 lists, deduplicated
printable reference

Every practice question shared during CKA prep, kept in its original wording. Part 1 is the original 51-question tiered bank. Part 2 is the later consolidated checklist, grouped by task type, with anything already covered in Part 1 removed.

Part 1 — Tiered bank (Q1–Q51)

Organized by how likely each is to appear on the real exam.

☐ Q1 Install a CNI that supports `kubectl` (, not apply), Installation CR NetworkPolicy — Calico via tigera-operator `create` with the cluster's pod CIDR.

☐ Q2 Reuse a `mariadb` (RWO, 250Mi), `spec.claimRef` if Released, retained PV `match` confirm with a new `storageClassName`, Bound before PVC `clear` applying the (MariaDB) — deployment. PVC

☐ Q3 Add a `busybox:stable` `tail` `/var/log/` via `emptyDir`. `restartPolicy:`). logging `<app>.log` Also know the Always sidecar native sidecar with a form (initContainers shared volume + —

☐ Q4 Create a `rancher.io/`, `volumeBindingMode:` , `is-` annotation, `csi-` StorageClass `local-path` `WaitForFirstConsumer` `default-` unset the `retain` and make it `class` old default. `sc` the default Variant: — provisioner

☐ Q5 Expose a container port and create a NodePort Service — port 80/ `nodePort` . http on nginx, selector matches pod labels, verify endpoints. 30080 Variant: fixed

☐ Q6 Migrate `web-` (hostname `gateway.web.k8s.local` , `tls.mode` , same an `gateway` HTTPS/ `Terminate` certificateR Ingress 443, + HTTPRoute to Gateway API — Gateway

☐ Q7 Create an

Ingress resource — echo in echo-sound host , example.org , / prefix, ingressClassName if present. Variant: expose the Deployment as NodePort port 8080, add

☐ Q8 HPA with scale-down stabilization — apache-server in autoscale , scaleDown.stabilizationWindowSeconds 50% 30 CPU, min 1 max 4,

☐ Q9 Choose the least-permissive NetworkPolicy — frontend → backend cross-namespace; check labels, prefer a namespaceSelector + podSelector combined from entry in one with a named port; don't touch existing deny-all policies.

☐ Q10 cert-manager explain CRDs and kubectl — kubectl get crd | grep cert-manager > ~/resources.yaml ; kubectl explain certificate.spec.subject > ~/subject.yaml

☐ Q11 PriorityClass and patch a Deployment — high-priority value = system-one * patch logger busybox-'s spec.template.spec.pr less than the highest user-defined class (ignore

☐

Q12 Helm: `https://argoproj.github.io/argo-helm`, `7.7.3`, `--set crds.install=false` on template and install both
Argo `argo-helm`
CD
without
CRDs
— repo

☐ **Q13** Split node resources across `describe`, subtract ~10–15% overhead then pods — WordPress, 3 `node` divide by 3 (~300m/600Mi); same replicas; check Allocatable requests for init and main containers.
via

☐ **Q14** Fix a broken control `--etcd-servers` in the apiserver `crictl + journalctl -u kubelet`.
plane after migration — single-node kubeadm with external etcd; the wrong port (2380 vs 2379);
debug via

☐ **Q15** Restrict nginx to `ssl_protocols TLSv1.3` only, `nginx-ssl`; add the `/etc/nginx/conf.d/` `Service` hosts rollout
TLSv1.3 via `ConfigMap` — `in config` IP to restart.

☐ **Q16** Taints and `node01`, schedule a `nodeSelector` / `nodeName` (a tolerations pod with a toleration
— taint matching toleration + alone
doesn't
force
placement).

☐ **Q17** Install `cri-dockerd`; `net.bridge.bridge-nf-call-iptables=1`, `net.ipv6.conf.all.forwarding=1`,
`sysctl`
`dockerd`
and set
`sysctl`
— `dpkg`
install,
enable/
start

- ☐ **Q18** Kustomize `nginx: , env=prod ; - , kustomize .`
`overlay — prod 1.27 label apply k preview build`
`overlay: with with`
`namespace, 3`
`replicas, image`
-
- ☐ **Q19** Helm `web , replicaCount=2 , upgrade a helm / helm .`
`install/ value, history get`
`upgrade/ rollback to values`
`rollback — revision 1;`
`release`
-
- ☐ **Q20** ConfigMap and `app- as APP_MODE , app- mounted /etc/ .`
`Secret config env secret read- secret`
`consumption — var only at`
-
- ☐ **Q21** Deployment rollback — break it `kubernetes.io/ ; -- is`
`with a bad image, roll back, record change - cause record deprecated).`
`the change-cause via annotation (`
-
- ☐ **Q22** Node affinity scheduling — required `disktype=ssd , preferred zone=a with a weight.`
-
- ☐ **Q23** Join a worker node `kubeadm token create -- , node02 , confirm`
`with kubeadm — print-join-command join Ready.`
-
- ☐ **Q24** Service with no endpoints — fix a label/selector mismatch or wrong targetPort without recreating the Deployment.
-
- ☐ **Q25** CoreDNS troubleshooting — scaled to 0 / broken `busybox: + nslookup .`
`Corefile / missing forward; fix it, test with 1.28`

☐ **Q26** Worker node NotReady — SSH in, usually `systemctl enable --now kubelet` .

☐ **Q27** `etcd -- / -- / -- / -- ; -- , update backup cacert cert key endpoints restore data- the and to a dir hostPath. restore new`
—
snapshot
with

☐ **Q28** RBAC user from CSR — create/approve the CSR, Role + `kubectl auth` .
RoleBinding for pods, test with `can-i --as`

☐ **Q29** NetworkPolicy: `allow- in fubar / echo , internal ,port`
allow a port `port- namespace 9000`
from one `from- or`
namespace — `namespace 9200/`
TCP.

☐ **Q30** NetworkPolicy: only port 80 `appychip / my- , internal pods port 80`
inside a namespace — `policy only, nothing external.`

☐ **Q31** NetworkPolicy with ipBlock, ingress and `except + selectors on 6379;`
egress — ingress from a CIDR with `egress to a CIDR on 5978.`

☐ **Q32** PVC, pod `pv- ( csi- , web- at /usr/ ,`
and PVC `volume hostpath- 10Mi, server share/ expand`
expansion `sc RWO), nginx/ to`
— `pod html 70Mi.`

☐ **Q33** Fix a pod that is not running — check `describe/events`: bad image, command,
`nodeSelector`, resources.

☐ **Q34** `kubeadm cluster upgrade` — upgrade control plane + worker, drain/uncordon each.

☐ **Q35** Ingress
with a

path to a service — `pong / ping in ing-internal , hello / hi at /hello on`

☐ [Q36](#) Extract log lines — `file-not-found` lines from pod `foo` to `/opt/KUTR00101/foo` .

☐ [Q37](#) Find the `kubectl top pod -l name=overloaded-cpu --sort-` .
highest-CPU `by=cpu > /opt/KUTR00401/KUTR00401.txt`
pod —

☐ [Q38](#) `hostPath` `app-` (2Gi, `/srv/`) `pv-` (100Mi, `/pv/`).
`PersistentVolume` `data` `ROX,` `app-` or `demo` `RWX,` `host-`
— `data` `data`

☐ [Q39](#) Identify control-plane component types — `process / static pod / pod`, for `kubelet`,
`apiserver`, `scheduler`, `controller-manager`, `etcd`, `DNS`.

☐ [Q40](#) Kubeconfig contexts — list all contexts, print the current one with and without `kubectl`.

☐ [Q41](#) Count Ready nodes without a `NoSchedule` taint — `/opt/KUSC00402/` .
write the count to `kusc00402.txt`

☐ [Q42](#) `JSONPath` `kubectl get nodes -o` .
query — `jsonpath='{.items[*].status.nodeInfo.osImage}' > /`
`root/all-nodes-os-info.txt`

☐ [Q43](#) Sort pods — by `metadata.creationTimestamp` and by `metadata.uid` .

☐ [Q44](#) DNS lookup for a pod `dns-` pod + `busybox:` , `/root/` .
and service — `resolver` service, `1.28` save `nginx.svc`
to

☐ [Q45](#) Rolling update with `change-cause` — `myproject nginx:1.16 → 1.17` , recorded.

☐ [Q46](#) Scale a Deployment — `presentation` to 3, or `guestbook` to 5.

☐

Q47 Multi-container pod — kucc8 (nginx+consul), kucc1 (redis+consul), pod-multi (nginx+busybox sleep 4800).
pod —

☐ **Q48** Pod with nodeSelector — nginx-kusc00401 with disk=ssd or disk=spinning .

☐ **Q49** Pod test- in custom (env=test , tier=backend) web- (tier=web-).
with pod or nginx app
labels
—

☐ **Q50** Pod with web- (busybox, sleep), securityContext.capabilities.add: .
SYS_TIME pod 3200 ["SYS_TIME"]
capability
—

☐ **Q51** ReplicaSet appychip (nginx: , 4 replicas); one pod already
alongside an 1.18 running; the RS adopts matching
existing pod — pods; total stays 4.

Part 2 — Consolidated checklist

The later master list, grouped by task type, with anything already listed in Part 1 removed.

Commands / queries

- ☐ List all pods sorted by name.
- ☐ List all pods sorted by created timestamp.
- ☐ List all pods sorted by age.
- ☐ Get the IP address of pod `nginx-dev` .
- ☐ List an nginx pod with custom columns `POD_NAME` and `POD_STATUS` .
- ☐ Get the image version in a pod without using `describe` .
- ☐ List pod name and namespace using JSONPath.
- ☐ List PVs sorted by capacity and save the output.
- ☐ Create an nginx pod and show different verbosity levels.

Pod tasks

- ☐ Create pod `kucc8` with containers `nginx`, `redis`, `memcached`.
- ☐ Create pod `non-persistent-redis` with an `emptyDir` volume.
- ☐ Create pod `pod1` on the master node only.
- ☐ Create static pod `webtool` on a node.
- ☐ Modify `hungry-`'s spec: add an init container `/workdir/`; the pod exits if the file `pod bear` that creates `calm.txt` is absent.
- ☐ Create pod `ready-if-service-ready`.
- ☐ Create pod `am-i-ready` and satisfy its readiness probe.
- ☐ Create multi-container pod `multi-container-playground`.
- ☐ Create a pod using ServiceAccount `secret-reader`.
- ☐ Create pod `p2-pod` with `nginx` + `busybox`.
- ☐ Create pod `secret-pod`.

Deployment tasks

- ☐ Create deployment `nginx-random`.
- ☐ Reconfigure deployment `front-end` and add a named port `http`.
- ☐ Restore a deleted MariaDB deployment.
- ☐ Create deployment `deploy-important`.

Service tasks

- ☐ Create NodePort service `front-end-svc` .
- ☐ Create service `nginx-random` .
- ☐ Create service `p2-service` .
- ☐ Troubleshoot a service with no endpoints.

Ingress tasks

- ☐ Create ingress `echo` .
- ☐ Create ingress `ping` .

NetworkPolicy tasks

- ☐ Select the correct NetworkPolicy from provided YAMLs for frontend/backend communication.
- ☐ Create `allow-port-from-namespace` .
- ☐ Create egress NetworkPolicy `np-backend` .

Secret / ConfigMap tasks

- ☐ Create secret `super-secret` .
- ☐ Mount a secret as a file.
- ☐ Inject a secret as an environment variable.
- ☐ ServiceAccount token access to the Kubernetes API.
- ☐ ConfigMap creation from literals / files / directories.

Storage tasks

- ☐ Restore MariaDB using a retained PV.
- ☐ Create PVC `mariadb` .
- ☐ Diagnose a Pending pod caused by a PVC.

DaemonSet tasks

- ☐ Create DaemonSet `ds-kusc00201` .
- ☐ Create DaemonSet `ds-important` .

HPA tasks

- ☐ Create HPA `apache-server` .

RBAC tasks

- ☐ Create a ServiceAccount.
- ☐ Create a Role.
- ☐ Create a RoleBinding.
- ☐ Create a ClusterRole.
- ☐ Create a ClusterRoleBinding.
- ☐ Verify permissions using `kubectl auth can-i` .

Node management

- ☐ Cordon a node.
- ☐ Drain a node.
- ☐ Uncordon a node.
- ☐ Add a label to a node.
- ☐ Add / remove taints.
- ☐ Set `ek8s-node-1` unavailable and reschedule its workloads.

kubeadm tasks

- ☐ Initialize a cluster using kubeadm.
- ☐ Configure `ik8s-master-0` as the control plane.
- ☐ Join `ik8s-node-0` .

etcd tasks

- ☐ Take an etcd backup.
- ☐ Verify the snapshot.
- ☐ Restore the snapshot.
- ☐ Identify etcd's key storage location.
- ☐ Find etcd's certificate expiration.
- ☐ Verify client certificate auth.
- ☐ Save a snapshot to `/etc/etcd-snapshot.db`.
- ☐ Fix a restored etcd hostPath / data-dir mismatch.

Certificate tasks

- ☐ Check the kubelet client certificate.
- ☐ Check the kubelet server certificate.
- ☐ Check expiry dates.
- ☐ Renew certificates.

DNS tasks

- ☐ Verify service DNS.
- ☐ Verify pod DNS.
- ☐ Save nslookup output.
- ☐ Troubleshoot CoreDNS.

CRI / containerd tasks

- ☐ Find a container with `crictl`.
- ☐ Save runtime info.
- ☐ Save container logs.
- ☐ Verify kube-proxy uses iptables.

Cluster discovery tasks

- ☐ Count master nodes.
- ☐ Count worker nodes.
- ☐ Find the Service CIDR.
- ☐ Identify the CNI plugin.
- ☐ Find the static pod naming suffix.

Event / monitoring tasks

- ☐ Save the latest cluster events.
- ☐ Delete the kube-proxy pod and observe events.
- ☐ Kill the kube-proxy container and compare events.

Troubleshooting tasks

- ☐ Troubleshoot a NotReady node.
- ☐ Troubleshoot kubelet.
- ☐ Troubleshoot containerd.
- ☐ Troubleshoot cert expiration.
- ☐ Troubleshoot a Pending pod.
- ☐ Troubleshoot CrashLoopBackOff.
- ☐ Troubleshoot ImagePullBackOff.
- ☐ Troubleshoot service connectivity.
- ☐ Troubleshoot a NetworkPolicy issue.
- ☐ Troubleshoot a DNS issue.
- ☐ Troubleshoot a control-plane failure.
- ☐ Troubleshoot an etcd failure.
- ☐ Troubleshoot a kube-proxy failure.
- ☐ Given a broken cluster, identify the failing node/service and restore health permanently.